

## QUARTILES

Quartiles are dividing points in the range of a set of data that give a clear picture of distribution. The median marks the center point, the upper quartile marks the midpoint between the median and the top of the distribution, and the lower quartile the midpoint between the median and the bottom. Estimates of quartiles can be found from a graph, or calculated precisely using formulas.

### Estimating quartiles

Quartiles can be estimated by reading values from a cumulative frequency graph (see p.213).

**Make a table with the data given** for range and frequency, and add up the cumulative frequency. Use this data to make a cumulative frequency graph, with cumulative frequency on the y axis, and range on the x axis.

| Range | Frequency | Cumulative frequency |
|-------|-----------|----------------------|
| 30–39 | 2         | 2                    |
| 40–49 | 3         | 5 (= 2+3)            |
| 50–59 | 4         | 9 (= 2+3+4)          |
| 60–69 | 6         | 15 (= 2+3+4+6)       |
| 70–79 | 5         | 20 (= 2+3+4+6+5)     |
| 80–89 | 4         | 24 (= 2+3+4+6+5+4)   |
| >90   | 3         | 27 (= 2+3+4+6+5+4+3) |

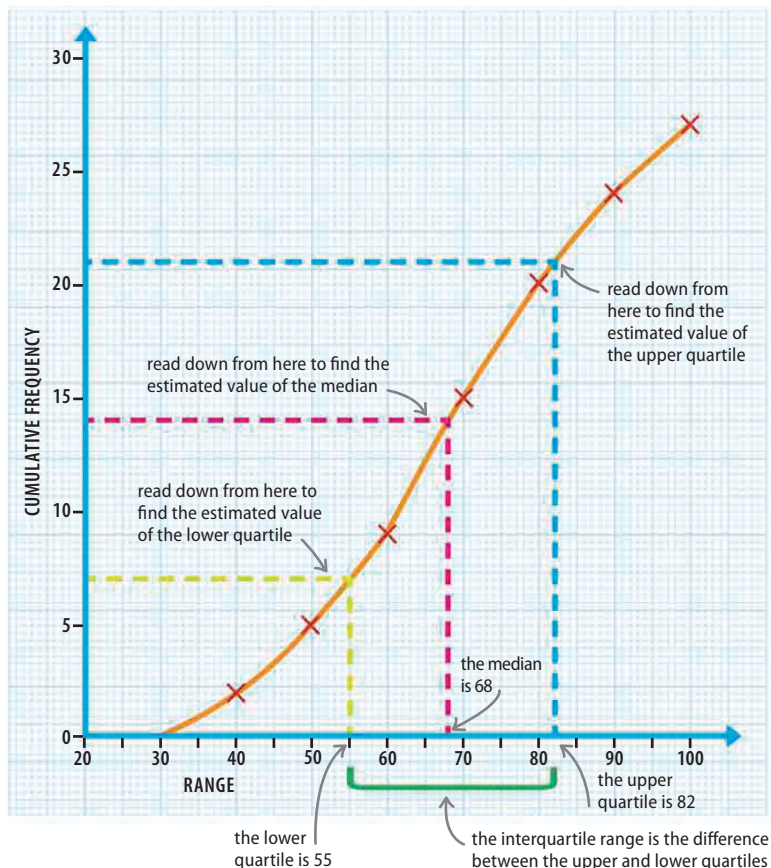
the sign means greater than

add each number to those before it to find cumulative frequency

**Divide the total cumulative frequency** by 4 (this will be the cumulative frequency of the last entry in the table), and use the result to divide the y axis into 4 parts.

total cumulative frequency  $\rightarrow \frac{27}{4} = 6.75$

divide the y axis into sections of this length



**Read across from the marks** and down to the x axis to find estimated values for the quartiles. These are only approximate values.

### Calculating quartiles

Exact values of quartiles can be found from a list of data. These formulas give the position of the quartiles and median in a list of data in ascending order, using the total number of data items in the list,  $n$ .

$n$  is the total number of values in the list

$$\frac{(n + 1)}{4}$$

△ **Lower quartile**

This shows the position of the lower quartile in a list of data.

$$\frac{(n + 1)}{2}$$

△ **Median**

This shows the position of the median in a list of data.

$$\frac{3(n + 1)}{4}$$

△ **Upper quartile**

This shows the position of the upper quartile in a list of data.